

Weakness

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Weakness refers to loss of muscle strength. That is, people cannot move a muscle normally despite trying as hard as they can. However, the term is often misused. Many people with normal muscle strength say they feel weak when the problem is fatigue or when their movement is limited because of pain or joint stiffness.

Muscle weakness can be a symptom of nervous system malfunction.

For a person to intentionally move a muscle (called a voluntary muscle contraction), the brain must generate a signal that travels a pathway from

- The brain
- Through nerve cells in the brain stem and spinal cord
- Through nerves from the spinal cord to the muscles (called peripheral nerves)
- Across the connection between nerve and muscle (called a neuromuscular junction)

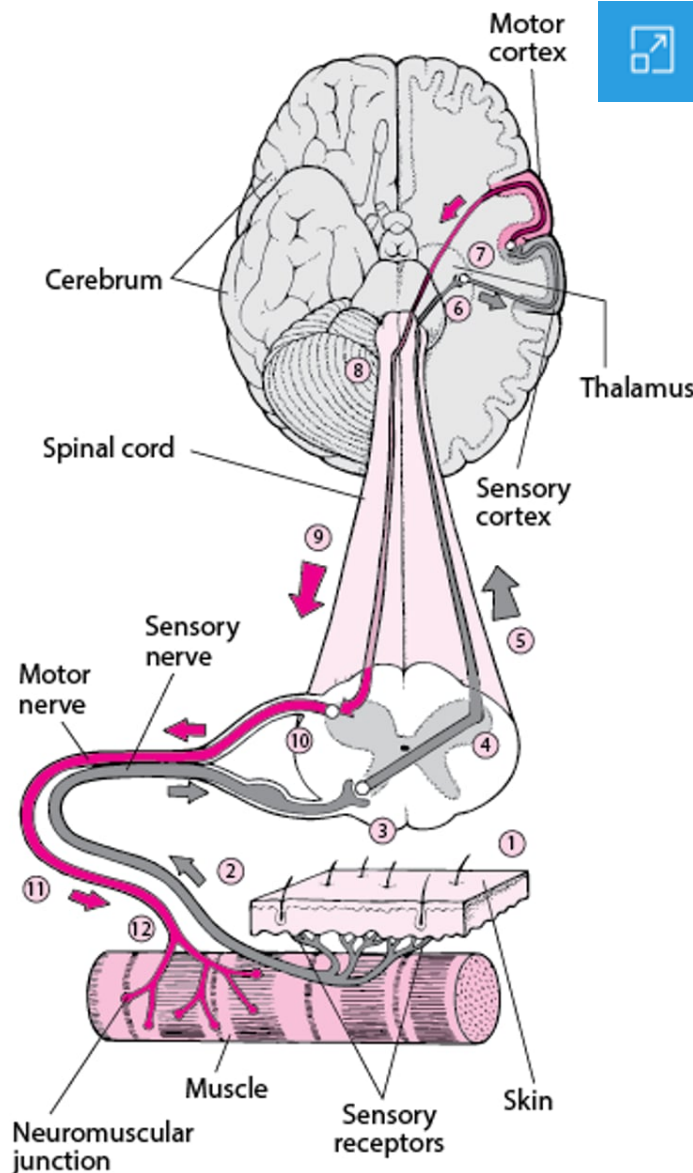
Also, the amount of muscle tissue must be normal, and the tissue must be able to contract in response to the signal from the nerves. Therefore, true weakness results only when one or more part of this pathway—brain, spinal cord, nerves, muscles, or the connections between them—is damaged or diseased.

Using the Brain to Move a Muscle

Moving a muscle usually involves communication between the muscle and the brain through nerves. The impetus to move a muscle may originate in the brain, as when a person consciously decides to move a muscle—for example, to pick up a book.

Or the impetus to move a muscle may originate with the senses. For example, special nerve endings in the skin (sensory receptors) enable people to sense pain or a change in temperature. This sensory information is sent to the brain, and the brain may send a message to the muscle about how to respond. This type of exchange involves 2 complex nerve pathways:

- The sensory nerve pathway to the brain
- The motor nerve pathway to the muscle



1. If sensory receptors in the skin detect pain or a change in temperature, they transmit an impulse (signal), which ultimately reaches the brain.
2. The impulse travels along a sensory nerve to the spinal cord.
3. The impulse crosses a synapse (the junction between two nerve cells) between the sensory nerve and a nerve cell in the spinal cord.
4. The impulse crosses from the nerve cell in the spinal cord to the opposite side of the spinal cord.
5. The impulse is sent up the spinal cord and through the brain stem to the thalamus, which is a processing center for sensory information, located deep in the brain.
6. The impulse crosses a synapse in the thalamus to nerve fibers that carry the impulse to the sensory cortex of the cerebrum (the area that receives and interprets information from sensory receptors).

7. The sensory cortex perceives the impulse. A person may then decide to initiate movement, which triggers the motor cortex (the area that plans, controls, and executes voluntary movements) to generate an impulse.
8. The nerve carrying the impulse crosses to the opposite side at the base of the brain.
9. The impulse is sent down the spinal cord.
10. The impulse crosses a synapse between the nerve fibers in the spinal cord and a motor nerve, which is located in the spinal cord.
11. The impulse travels out of the spinal cord along the length of the motor nerve.
12. At the neuromuscular junction (where nerves connect to muscles), the impulse crosses from the motor nerve to receptors on the motor end plate of the muscle, where the impulse stimulates the muscle to move.

If the sensation occurs suddenly and is severe (as when stepping on a sharp rock or picking up a cup of very hot coffee), the impulse may travel to the spinal cord and directly back to the motor nerve, bypassing the brain. The result is a quick response of a muscle—by immediately withdrawing from whatever is causing the pain. This response is called a spinal reflex.

Weakness may develop suddenly or gradually. Weakness may affect all of the muscles in the body (called generalized weakness) or only one part of the body. For example, depending on where the spinal cord is damaged, spinal cord disorders may cause weakness only of the legs.

Symptoms depend on which muscles are affected. For example, when weakness affects muscles of the chest, people may have difficulty breathing. When weakness affects muscles that control the eyes, people may have double vision.

Complete muscle weakness causes paralysis. People may have other symptoms depending on what is causing the weakness. Weakness is often accompanied by abnormalities in sensation, such as tingling, a pins-and-needles sensation, and numbness.

Causes of Weakness

Because malfunction in the same part of the signal pathway causes similar symptoms regardless of cause, the many causes of muscle weakness are usually grouped by the location of the cause (see table [Some Causes and Features of Muscle Weakness](#)). That is, causes are grouped as those that affect the brain, spinal cord, peripheral nerves, muscles, or connections between nerves and muscles. However, some disorders affect more than one location.

Common causes

Causes differ depending on whether weakness is generalized or affects only specific muscles.

For **generalized weakness**, the most common causes are:

- A decrease in general physical fitness (called deconditioning), which may result from illness and/or a decrease in physical reserves (frailty), such as muscle mass, bone density, and the heart's and lungs' ability to function, especially in older adults
- Loss of muscle tissue (wasting, or atrophy) due to long periods of inactivity or bed rest, as occurs in an intensive care unit (ICU)
- Damage to nerves due to a severe illness or injury, such as severe or extensive burns
- Certain conditions that affect muscle, such as a low level of potassium ([hypokalemia](#)), consumption of too much alcohol, or use of steroids (sometimes called glucocorticoids or corticosteroids)
- Medications used to paralyze muscles—for example, to keep people from moving during surgery or while on a ventilator

For **weakness in specific muscles**, the most common causes are:

- [Strokes](#) (the most common cause of weakness affecting one side of the body)
- Nerve damage, as occurs in [carpal tunnel syndrome](#) or results from injury
- A [ruptured or herniated disk](#) in the spine
- [Pressure on \(compression of\) the spinal cord](#), as can result from cancer that has spread to the spinal cord
- [Multiple sclerosis](#)

Less common causes

Many other conditions sometimes cause weakness (see table

~~Some Causes and Features of Muscle Weakness~~). For example, electrolyte abnormalities (such as a low level of [magnesium](#) or [calcium](#)) can cause weakness that sometimes comes and goes, as well as [muscle cramping](#) and twitches.

In people with a seizure disorder, one side of the body may become weak after a seizure stops (called Todd paralysis). The weakness usually subsides over several hours.

Low blood sugar ([hypoglycemia](#)) can also cause weakness, which resolves when hypoglycemia is treated.

Fatigue

Many people report weakness when their problem is actually [fatigue](#). Common causes of fatigue include a severe illness, cancer, a chronic infection (such as [HIV infection](#), [hepatitis](#), or [mononucleosis](#)), [heart failure](#), [kidney failure](#), [liver failure](#), [anemia](#), [myalgic encephalomyelitis/chronic fatigue syndrome](#), [fibromyalgia](#), and mood disorders (such as [depression](#)).

Multiple sclerosis can cause fatigue that increases when people are exposed to heat and humidity.

Evaluation of Weakness

First, doctors try to determine whether people are weak or simply tired. If people are weak, doctors then determine whether the weakness is severe enough or worsening quickly enough to be life threatening. Doctors also try to identify the cause.

Warning signs

In people with weakness, the following symptoms are cause for concern:

- Weakness that becomes severe over a few days or less
- Difficulty breathing
- Difficulty raising the head while lying down
- Difficulty chewing, talking, or swallowing
- Loss of the ability to walk

When to see a doctor

People who have any warning sign should go to an emergency department immediately. Immediate medical attention is crucial because weakness accompanied by a warning sign can worsen quickly and cause permanent disability or be fatal.

If people have [symptoms suggesting a stroke](#) (see table [Some Causes and Features of Muscle Weakness](#)), they should seek medical attention immediately because early treatment can help limit loss of function and sensation.

People without warning signs should call their doctor. The doctor can decide how quickly they need to be seen based on their symptoms and other disorders they have.

If the weakness worsens gradually (over months to years), people should discuss the problem with their doctor at their next visit.

What the doctor does

Doctors first ask questions about the person's symptoms and [medical history](#). Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause and the tests that may need to be done (see table [Some Causes and Features of Muscle Weakness](#)).

Doctors ask people to describe in detail what they are experiencing as weakness. Doctors ask:

- When the weakness began
- Whether it began suddenly or gradually
- Whether it is constant or is worsening
- Whether it varies throughout the day
- Which muscles are affected
- Whether and how the weakness affects the ability to do certain activities, such as breathing, brushing their teeth or hair, speaking, swallowing, standing up from a seated position, climbing stairs, and walking
- Whether they have other symptoms that indicate malfunction of the nervous system, such as speech or vision problems, loss of sensation or memory, or seizures
- Whether any activity or condition (such as heat or repetitive use of a muscle) makes the weakness worse

What seems to be sudden weakness is sometimes gradual weakness, but people do not notice it until they can no longer do something, such as walking or tying their shoes.

Based on the description of weakness, doctors can often identify the most likely causes, as for the following:

- A muscle disorder: Weakness beginning in the hips and thighs or the shoulders (that is, people have difficulty standing up or lifting their arms overhead) and no effect on sensation
- A peripheral nerve disorder: Weakness beginning in the hands and feet (that is, people have difficulty lifting a cup, writing, or stepping over a curb) and loss of sensation

Doctors also ask about other symptoms, which may suggest one or more possible causes. For example, if people with back pain and a history of cancer report weakness in a leg, the cause may be cancer that has spread and put pressure on the spinal cord.

Doctors ask about conditions that increase the risk of disorders that cause weakness. For example, they ask people whether they have high blood pressure and atherosclerosis (which increase the risk of stroke) or have traveled to an area where Lyme disease is common.

People are asked about symptoms that suggest [fatigue](#) or another problem, rather than true muscle weakness. Fatigue tends to cause more general symptoms than true muscle weakness, and fatigue does not follow a particular pattern. That is, it is present all the time and affects the whole body. People with true muscle weakness often report difficulty doing specific tasks, and the weakness follows a pattern (for example, becomes worse after walking).

Doctors ask about recent or current disorders that commonly cause fatigue, such as any recent severe illness or a mood disorder (such as [depression](#)).

Doctors ask about past and current use of medications, alcohol, and illicit drugs.

Whether family members have had similar symptoms can help doctors determine whether the cause is hereditary.

During the physical examination, doctors focus on the nervous system ([neurologic examination](#)) and muscles.

Doctors observe how the person walks. How people walk may suggest the disorder that is causing symptoms or its location. For example, if people drag a leg, do not swing one arm as much as the other when walking, or both, their symptoms may be

caused by a stroke. Doctors also check for other signs that the nervous system is malfunctioning, such as loss of coordination or sensation.

Cranial nerves (which connect the brain with the eyes, ears, face, and various other parts of the body) are tested—for example, by checking eye movements, the ability to speak clearly, and the ability to rotate the head (see table

[Testing Cranial Nerves](#)

Muscles are checked for size and unusual unintended movements (such as involuntary twitches and shaking). Doctors note how smoothly muscles move and whether there is involuntary resistance to movement (detected when doctors try to move a muscle that they have asked the person to relax).

Reflexes are checked. Reflexes are automatic responses to a stimulus. For example, doctors test the knee jerk reflex by gently tapping the muscle tendon below the kneecap with a rubber hammer. Normally, the knee then jerks involuntarily. This evaluation helps doctors identify which part of the nervous system is probably affected, as for the following:

- The brain or spinal cord: If reflexes are very easy to trigger and are very strong
- The nerves: If reflexes are hard to trigger and are slow or absent

Muscle strength is tested by asking the person to push or pull against resistance or to do maneuvers that require strength, such as walking on the heels and tiptoes or standing up.

A general physical examination is done to look for other symptoms that may suggest a cause, such as shortness of breath (possibly caused by a heart or lung disorder).

Generally, if the history and physical examination do not detect specific abnormalities that suggest a brain, spinal cord, nerve, or muscle disorder, the cause is likely to be fatigue.



Some Causes and Features of Muscle Weakness

Cause	Common Features*	Diagnostic Approach†
Brain disorders		
Brain tumors	Headaches, personality changes, confusion, difficulty concentrating, drowsiness, loss of balance and coordination, and paralysis or numbness Sometimes seizures	MRI or CT of the brain
Multiple sclerosis (affects the brain, spinal cord or both)	Usually other symptoms of nervous system malfunction (such as loss of sensation, loss of coordination, and vision problems) Weakness that: • Tends to come and go • Sometimes affects different parts of the body • Is worse in hot weather	MRI of the brain and sometimes of the spinal cord Sometimes a spinal tap (lumbar puncture) Sometimes evoked responses
Stroke	Symptoms that occur suddenly: • Weakness or paralysis, usually on one side of the body	CT or MRI of the brain



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